

CareerSync AI

A Multi-Agent Agentic System for End-to-End & Personalised Career Transition Support.

SDGs
We aim for

8 DECENT WORK AND ECONOMIC GROWTH



4 QUALITY EDUCATION



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Introduction

Career transition is a structurally underserved problem. While professional networking platforms aggregate job listings and skill taxonomies, they offer no **personalised** reasoning layer. A software engineer doesn't know how their skills map to healthcare IT, what to learn in 30 days, or how to perform in a domain-specific interview they have never attempted before.

Existing AI-assisted tools treat career guidance as a **retrieval** problem. We reframe it as an **agentic reasoning** problem..

Objective

To design, implement, and evaluate a **multi-agent AI** system that enables end-to-end **career transition** support.

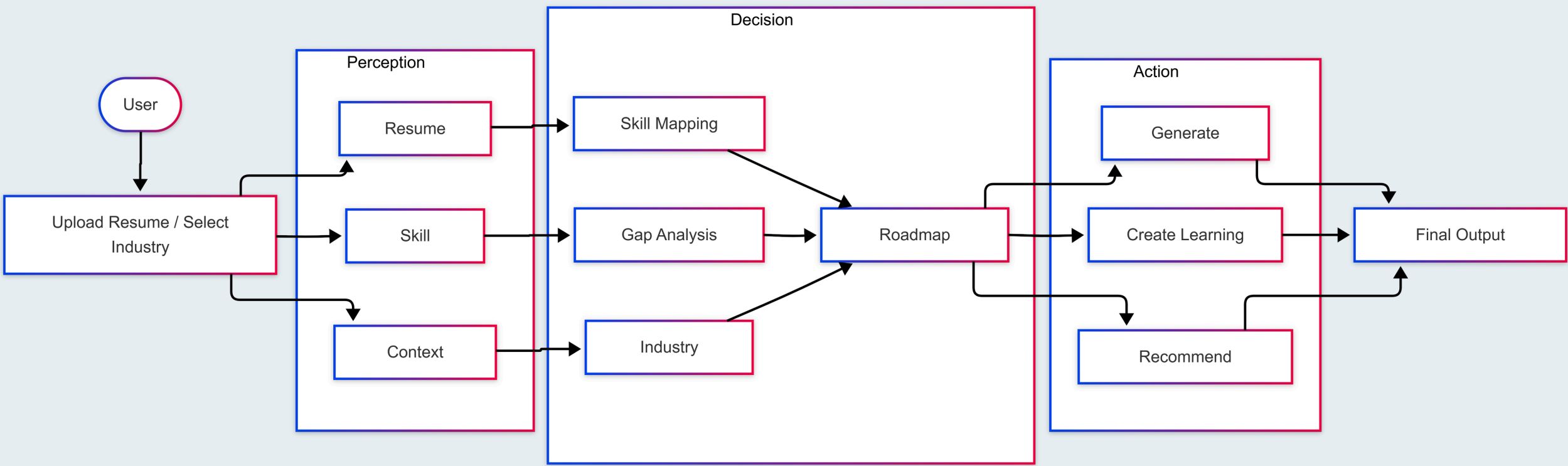
Methodology

The system accepts a raw resume, a target industry, and a preparation timeline, then autonomously executes a five-stage analysis pipeline, extracting skills, mapping transferable competencies, identifying gaps, generating a phased learning roadmap, and simulating a progressively difficult interview

Results

The system successfully executed the full Pivot Mode pipeline, skill extraction through interview report, using five specialised agents designed with distinct reasoning responsibilities:

1. **Profile Agent** (skill extraction)
2. **Mapping Agent** (transferability scoring with High/Medium/Low relevance)
3. **Roadmap Agent** (4-phase timeline decomposition)
4. **Dashboard Agent** (confidence scoring and job matching)
5. **Interview Agent** (question generation and 4-dimension evaluation: Relevance, Clarity, Depth, Technical Accuracy).



Analysis

- The Profile Agent correctly extracts 6–9 skills per resume
- The Mapping Agent assigns relevance scores and identifies 3–6 domain-specific gaps per transition.
- The Roadmap Agent consistently returned 4-phase plans, with 4 topics per phase and resource links to Coursera, YouTube, freeCodeCamp, Medium / Google per phase.
- The Interview Agent generates 5 unique, non-repeating questions per session with increasing difficulty; evaluates unstructured responses as average and STAR-method structured responses as above average.

Conclusion

CareerSync AI demonstrates that a coordinated multi-agent LLM pipeline can deliver end-to-end career transition intelligence.

The system's agent decomposition strategy, structured JSON output enforcement, and on-demand interview generation collectively address the core failure modes of single-prompt career AI.

The penalised confidence score formula produces readiness signals that align with intuitive domain proximity assessments.

